



## **OHIO RIVER TRAIL FEASIBILITY STUDY**

**June 17, 2010**

### **Ohio River Trail Proposed Route - Operation and Maintenance Program**

Maintenance is a key facet of all trail management programs. Effective maintenance provides trail users with a clean, safe and enjoyable trail experience while protecting trail ownership from unwanted liability claims. Proper maintenance is also a means to keep the cost of managing the trail in check over the long-term.

#### **General Inspections**

General trail inspections provide management with an overall sense of the condition of the trail and allow for planning of upcoming maintenance needs and projects. Trail inspections can occur regularly throughout the year but a spring trail inspection is of particular importance because of the potential problems created by 3-4 months of winter weather. Trail ambassador and friends of the trail groups are particularly valuable inspection resource because of their frequency of use on the trail. These groups can be advised what to look for and a reporting system established to relay problems to the appropriate management entity. It is also good practice to keep records of inspections and maintenance activities.

#### **Drainage Inspections**

The inspection of drainage facilities which include ditches, swales, culverts and cross pipes, is of particular importance in the overall health of the trail. When functioning properly, they ensure water is directed and handled accordingly, preventing unwanted erosion of the trail. Inspections facilitate that the drainage system is operating correctly and that impediments can be removed before they cause problems. These should be conducted in the spring and again in the late fall and can be performed by a variety of parties, including volunteers.

#### **Surface Treatment**

Treating the surface of trail is dependent on the type of surface. Crushed stone trails present more maintenance concerns than asphalt trails. Crushed stone trails require some form of grading or dragging on at least an annual basis, sometimes more frequently depending on usage patterns. This can be accomplished with machinery designed for the task or with a drag pulled behind a truck (many regional trail groups have had success in design in "homemade or machine fabricated drags). Asphalt trail surfaces do not require dragging or grading

A wooden deck surface, such as that proposed for the boardwalk area between CSX property and the Ohio River also requires specific maintenance. The cost for the care of the surface can be mitigated drastically by using a "trex" or recycled plastic lumber deck. Traditional treated wood decking surfaces require preservative treatment every 2-4 years and the costs of the work and material soon exceed the

replacement cost of the wood. The surface of decks should be inspected regularly to ensure that fasteners have not worked themselves loose, which pose potential risks to trail users. This can be easily accomplished by municipal staff or volunteer crews.

### **Mowing**

Mowing, where vegetative shoulders exist along the side the trail, is important to provide for safe, clear recovery zones and also for aesthetic purposes. Mowing shoulders also helps to slow the encroachment of vegetation into the trail surface material, particularly a problem with crushed stone surfaces. In most cases a four to five foot swath is an appropriate mowing width. In heavily used areas, close to access points, wider mowing swaths can be desirable. Mowing can be accomplished through contracted services, municipal cooperation or by the trail group's volunteer corps. Trail groups also have employed a system of purchasing the equipment and contracting with an operator or coordinating the volunteers. Mowing cycles generally are 1-2 months apart, resulting in mowing requirements of 3-5 times annually. Mowing at parking access areas is an important part of vegetation management. These are often the first area visitors see and a clean, well-mowed area presents a good first impression. In addition to mowing, hand trimming is also recommended. These areas can be maintained by volunteer labor groups.

### **Spraying**

Both asphalt and crushed stone trails can require some herbicidal spraying as part of the maintenance program. Spraying helps to prevent parallel vegetation encroachment that, over time, can considerably decrease the width of the trail. It also helps to control vegetation infiltration into the trail surface itself. On crushed stone trails, a spraying/dragging program is particularly effective at managing infiltration. Keep in mind that the application of herbicides should only be conducted by a licensed applicator. Many municipal recreation or public works departments keep someone on staff licensed to spray. Since spraying is usually only required once during the year, late spring preferably, stay open to the possibility of co-opping.

### **Tree trimming**

As a maintenance operation, tree trimming helps to maintain vertical clearances and also to catch potential problems limbs/trees before they fall to the ground. Trimming should be conducted initially in the spring to remove tree falls and limb hangers. It is also a good idea to reevaluate the trimming in the late summer or early fall when the trees are vegetated so the effects on vertical clearances can be accurately judged.

## **Sweeping**

Both asphalt and crushed stone surfaces can require sweeping, generally performed in the spring and mid-fall, to remove vegetative debris from the trail. This can be accomplished with machinery designed for this purpose or through cheaper means such as placing individuals on a pickup bed with leaf blowers.

## **Litter and Graffiti Control**

The enforcement of graffiti and litter ordinances will be the responsibility of the relevant municipal police force. Litter control can be accomplished by educating volunteer groups and trail ambassadors to remove litter when they encounter it. Routine monitoring and patrolling by trail users and local support groups will minimize this problem as well.

## **Law Enforcement/Security**

In general, law enforcement will be handled by the local law enforcement agency, with the state police serving as backup. When there is no local agency available; the state police will provide primary law enforcement services.

## **EMS**

When emergencies arise on the trail, the local emergency response networks will generally manage the responses. Access to the local networks is through dialing 911. It is important that coordination with the appropriate EMS providers take place to ensure that access is attainable to areas of the trail system where vehicular use is restricted. It is also important to coordinate with County EMS to develop mapping as trail sections are developed and opened to aid in emergency response. The trail, along with mileposts, should be included in the County GIS inventory used by 911 for emergency dispatching.

## **Vehicular Use**

Although strongly discouraged, there will be times that vehicular use of the trail by others is a necessity. A vehicular use policy should be established that outlines the request process and guidelines for use of the trail. The trail group could request a fee for the privilege.

## **Signage**

A simple standardized signage system should be designed and implemented to deliver clear, concise messaging and promote continuity of the trail system. An effective signage program would include trailblazer signs defining the ORT Trail route as well as connection to neighboring feeder trails. Trail directional signs should provide information related to the location of support services, major destinations and attraction. Mile markers, regulatory signs and interpretive signs for historic and cultural features as provide valuable support information to trail users. In addition to its primary purpose, signage can be an important tool in helping to brand the trail. In some cases, the Federal Department of Transportation, Highways Administration, Manual of Uniform Traffic Control Devices (MUTCD) signage will be required which have clear, established guidelines for use and generally are

used to alert vehicular traffic of at-grade crossings. Signage should be replaced when damaged or faded.

### **Friends Group**

As the trail is developed, the continued growth and involvement of the Ohio River Trail Council will be an important asset to the long-term care of the trail. The development of an ORTC maintenance committee could spearhead and coordinate the overall care of the trail, where it is likely that a conglomeration of different parties will be participating in the maintenance of the trail. If municipal partners are involved, they should be offered seats on the committee. The ORTC should also develop a volunteer based "trail care crew" whose role would be to provide maintenance assistance and help on special projects matched to skill sets of the members.

### **Trail Ambassador Program**

The ORTC should implement a Trail Ambassador program once sections of trail come on-line. Many of the regional trail groups have established programs that could be copied and modified to meet the needs of the ORTC. Outside of providing a friendly, helpful face to the trail, ambassadors provide an important monitoring function. Often, ambassadors spend as much or more time on the trail than the managing entity and are more in tune with state of the trail.